

Section 4: Broader Impacts

Outline for Science Justification

Abstract: Summarizes the entire proposal

Section 1: Facts and summary of Sections 2-4

Section 2: The Problem/Motivation/Story

Section 3: The Goal & Strategy. How you will use “key component” + “Target” to solve the Problem

Section 4: Importance of the solution to the subfield & Broader Impacts (legacy value)

Broader Impacts and Addressing the Funding Call

Robinson, NASA XRP Abstract

solar system science. Finally, the proposed work focuses on “[d]etection of exoplanets and/or confirmation of exoplanet candidates” and, thus, falls within the purview of NASA’s Exoplanets Research Program. With regards to NASA’s Exoplanet Exploration Program’s 2025 Science Gaps List, the proposed work addresses the needed capabilities of SCI-06 (“Yield Estimation for Exoplanet Direct Imaging Missions”) focused on “[i]mproved treatment of observation scheduling and mission rule optimization.” Lastly, the proposed work enables the development of HWO and, thus, aligns with NASA’s 2025—2026 Science Plan and its commitment to the Decadal Surveys as well as the elements of NASA’s Science Strategy 1.1 that emphasize the search for life.

Broader Impacts and Addressing the Funding Call

Holt: NASA ROSES Abstract

Relevance to Solar System Workings:

This proposal is relevant to Solar System Workings because the program specifically supports #field studies of terrestrial analogs of planetary environments," which is the major component of our proposed work, providing the needed basis to answer fundamental processes regarding volatiles, climate and the cryosphere of Mars.

Our project addresses the following topical area under Surfaces and Interiors, as described in the announcement: [Evolution and modification of surfaces. Characterize and understand the chemical, mineralogical, and physical features of planetary surfaces (such as geologic formations and impact craters) and fluid inventories that interact with the surface (including hydrospheres, cryospheres, atmospheres, and other volatile reservoirs). Develop theoretical and experimental bases for understanding these features in the context of the varying conditions through time after formation.]

This project will also address basic questions about DCG such as debris thickness and variability, which is important for potential access in future exploration and in-situ resource utilization on Mars, both NASA priorities.

This proposal would not be appropriate to MDAP because it includes a major fieldwork component.

The Ancient Star Formation History of the Isolated, Quenched Ultra-faint Dwarf Galaxy Tucana B

We request F090W+F150W NIRCам imaging of Tucana B, a newly discovered, isolated ultra-faint dwarf galaxy beyond the edge of the Local Group ($M_V = -6.9$; $D = 1.4$ Mpc).

Tucana B is a unique ultra-faint, as it has likely never had an encounter with the Milky Way's hot halo or tidal field, yet its ground-based observations only display an old (~ 13 Gyr), metal poor ($[Fe/H] = -2$) stellar population, and it lacks any HI gas with which to

make new stars -- it is quenched. Reionization is expected to quench many of the smallest galaxies at early times, heating their gas reservoirs and stopping any subsequent star formation.

Observations of Tucana B down to the oldest main sequence turnoff will be a definitive test of the role that reionization plays, identifying whether early star formation stopped around the reionization epoch, or continued up until more recent times, as

is seen in more massive field dwarfs. Only JWST can perform these observations well outside the Milky Way's virial radius, where ultra-faint dwarf galaxies are unaffected by ram pressure and tidal forces. Tucana B is

currently the only target that can test this prediction of modern galaxy formation theory.

We have designed our observations with potential, future proper motion measurements in mind, which could further elucidate the origins and history of Tucana B with respect to the Local Group.

Facts

Importance of Facts

Problem

Proposal

Strategy - Soln

Importance of Solution

Broader Impact

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WHAT Prediction?